Ankit Billa

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Relevant Work Experience

Sauron Industries Inc. San Francisco, CA
Senior Perception Engineer *May 2025 – Present*

- Lead a team of 3 Perception Engineers; own the architecture and technical roadmap of the proprietary multi-camera, multi-sensor (RGB + LiDAR) perception stack powering real-time multi-person-vehicle tracking for an Autonomous Home Security system protecting high-risk, high-net-worth clientele.
- Designed the end-to-end "Human-in-the-Loop" threat-alerting pipeline, integrating real-time multi-sensor detection feeds with a custom-built Alert Manager, achieving sub-second event-to-operator latency for threat assessment by security operators.
- Deployed Foundation Models (DINOv3) and custom 3D Object Detection algorithms onto edge compute platforms (NVIDIA Jetson Orin, Thor) via post-training quantization (PTQ) with TensorRT and ONNX.
- Built the end-to-end pre-deployment tooling suite: a web application generating high-fidelity 3D property reconstructions from drone footage via photogrammetry and point-cloud mesh reconstruction, plus sensor-pod placement simulation with automatic sensor-view overlap analysis — enabling installation plans to be visualized and validated in a single first site visit.

Dragonfruit AI Inc. San Francisco, CA
Computer Vision Engineer (Machine Learning Platform Team) *May 2023 – Dec 2024*

- Developed & optimized a microservices pipeline for Multi-Camera Multi-Entity tracking, increasing pipeline throughput by 15%.
- Developed a novel entity re-identification algorithm combining graph-based matching & deep learning, improving accuracy by 25%.
- Led multiple model releases for robust entity detection & clustering, deployed over 40+ locations, overseeing a team of 4 engineers.
- Optimized legacy codebase for both on-premises and cloud deployments, enabling real-time insights from video streams.

Standard AI Inc. San Francisco, CA
Machine Learning Engineer (Object Tracking Team)(Co-Op) *Jun 2022 – Dec 2022*

- Implemented a novel algorithm for multi-view 3D reconstructions & multi-object tracking, with simultaneous space-time association, deployed in production across 100+ retail stores.
- Optimized the graph-based algorithm for handling higher number of camera channels, improving multi-camera multi-person tracking accuracy by 40% and processing speed by 15%.

Karlsruhe Institute of Technology | Fraunhofer IOSB Karlsruhe, Baden-Württemberg, Germany
Research Engineer (Vision & Fusion Laboratory) *Jan 2020 – July 2021*

- Spearheaded a novel data embedding technique for document fingerprint synthesis using generative & deep learning methods.
- Developed custom architecture combining GANs & Variational Autoencoders, achieving 98% accuracy in fingerprinting.
- Created and curated a large-scale dataset of 55,000 high-resolution paper texture images for generative model training.
- Benchmarked the model's performance against traditional image processing and wave filtering techniques, demonstrating superior generative capabilities on the proposed custom dataset.

Research

Off-Road Terrain Classification on a Novel RGB-D Dataset | GRASP Lab, University of Pennsylvania

- Building a large-scale in-the-wild RGB-D benchmark for depth-assisted semantic segmentation of off-road terrain, advised by Prof. Jianbo Shi; benchmarking ∼40 models spanning monocular depth estimation, semantic segmentation, RGB-D fusion, and foundation-model backbones.

Egocentric Motion Tracking for 3D Scene Interactions | GRASP Lab, University of Pennsylvania

- Engineered a pipeline for detecting three degrees of hand-object interactions in video streams, enabling precise contact point detection.
- Developed a pseudo-labeling algorithm for 3D contact point identification, reducing manual annotation time by 65% while maintaining 88% labeling accuracy compared to human experts, streamlining the creation of large-scale datasets for graphics research.
- Formulated a multi-modal deep learning architecture integrating RGB images, depth maps, and IMU data to improve 3D interaction tracking accuracy by 25% over single-modality approaches.

Advanced 3D-to-2D Correspondence Mapping for Real-time Subject Tracking

- Developed an algorithm for mapping 3D character meshes onto 2D video subjects via scene flow and pose estimation, achieving real-time tracking at 30FPS with a 40% reduction in mesh-to-subject alignment errors.

Skills

Programming: C, C++, Python, MySQL, OpenGL, GLSL, CUDA
Libraries & Frameworks: numpy, pandas, scikit-learn, Tensorflow, PyTorch, Keras, OpenCV, CVAT
MLOps & Deployment: Docker, Kubernetes, CI/CD Pipelines, Edge AI deployment, Model optimization & quantization
Cloud Platforms & Version Control: AWS, Google Cloud Platform, Azure, Git, Github, Gitlab
Deep Learning & Computer Vision Expertise: Advanced Object Detection & Tracking, Multi-Camera Calibration & 3D Reconstruction, Depth Estimation & SLAM, Semantic Segmentation & Instance Segmentation, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformer architectures, Generative Adversarial Networks (GANs), Reinforcement Learning, Neural Radiance Fields (NeRF), Pose Estimation & Motion Tracking

Education

University of Pennsylvania Philadelphia, PA
M.S.E. in Computer and Information Science *2021 - 2023*

Relevant Coursework: Advanced Machine Perception, Computer Vision & Computational Photography, Interactive Computer Graphics, Internet & Web Systems, Distributed Software Systems, Applied Machine Learning, Brain Computer Interfacing, 3D Machine Perception, Analysis & Design of Algorithms